

HARSH SAW

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SUMMARY

Full-stack software engineer with 2+ years of internship experience building production web applications, developer tools, and distributed real-time systems. Built a browser-based IDE with containerized sandboxes; shipped desktop apps, a 30K-user multi-channel messaging platform, and an HLS video pipeline — across TypeScript/React, Go, and Python stacks.

EDUCATION

Bachelor of Computer Information Systems
Okanagan College

Expected Dec 2026
Kelowna, BC, Canada

EXPERIENCE

- OmMuse** Seattle, US (Remote)
Software Engineer Intern *Nov 2025 – Present*
- Engineered a cross-platform desktop uploader (Electron) with persistent folder-sync (chokidar + SQLite WAL), parallelized **S3 multipart uploads**, **OAuth 2.0 PKCE** auth flow, and a typed IPC bridge exposing 70+ **gRPC** methods to the **TypeScript/React** renderer.
 - Shipped features across a polyglot monorepo (**Go gRPC/protobuf, TypeScript/React, Python**) unified by Bazel, owning full AWS deployment — S3, SQS, ECR, Docker, Pulumi IaC, Envoy proxy, and GitHub Actions CI/CD.
 - Built HubSpot CRM sync in **Go** for a **12K+ user** platform (18K+ tracks, 67 GB audio); architecting **RAG pipeline** (MongoDB Vector Store, AWS Bedrock) enabling natural language queries across projects and collaborators.
 - Designed async batch worker (sync.Map dedup, 30s flush, 100-user chunks) syncing live metrics across 8 API trigger points with zero data loss under concurrent load.
- MoreThinks Solutions Ltd.** Burnaby, BC
Full-Stack Developer Intern *Jun 2025 – Sept 2025*
- Developed full-stack features with **Qwik.js** and SurrealDB in a monorepo (AWS EKS CI/CD); engineered vector embedding pipelines consumed by ML recommendation services.
- Bwisher Ltd.** Remote
Software Engineer Intern *Dec 2024 – Jun 2025*
- Architected a multi-channel marketing platform using **Apache Kafka** + BullMQ, orchestrating personalized SMS, email, and WhatsApp campaigns across **30K+ users** while feeding real-time CRM and analytics pipelines.
 - Designed idempotent **NestJS** + **FastAPI** microservices for payment/credit workflows with **retry and exponential backoff**; owned Jenkins, Kubernetes, Prometheus, and Grafana deployment with feature-flag rollouts.
 - Built FastAPI demand forecasting services generating ML-driven segmented targeting for the Kafka campaign pipeline; eliminated duplicate transactions in high-stakes financial operations.
- Jythu Ltd.** Remote
Full-Stack Developer Intern *Jan 2024 – Dec 2024*
- Shipped two production **React Native** apps (LMS + admin) with Node.js backends serving 2K+ learners; built FFmpeg HLS streaming pipeline (S3/CloudFront + SQS/Lambda/SNS) cutting seek time **30s → 1.5s**, bandwidth **–70%**; provisioned AWS with **Terraform** + GitHub Actions CI/CD.

PROJECTS

- CodeExpo — Browser-Based IDE & Sandbox** | *React, Node.js, TypeScript, Docker, Nginx, xterm.js, Socket.IO* [Demo](#) | [GitHub](#)
- Full-stack browser IDE with live terminal (**xterm.js** over WebSocket), real-time code execution in browser preview pane, and **Socket.IO** file sync; **Docker** multi-stage sandboxes with Nginx reverse proxy, isolated per-session execution contexts.
- Document Intelligence Platform (RAG)** | *FastAPI, LangChain, Qdrant, Groq, Gemini, AWS ECS Fargate* [Demo](#) | [GitHub](#)
- Multi-tenant RAG platform with session-isolated Qdrant indexes, LangChain provider failover (Groq→Gemini), streaming retrieval, SHA-256 idempotent ingestion; deployed to AWS ECS Fargate with GitHub Actions CI/CD.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Go (Golang), Bash
Frontend & Mobile: React, Next.js, React Native, Qwik.js, Electron, xterm.js, Socket.IO
Backend & Messaging: NestJS, FastAPI, Express, Fastify, GraphQL, gRPC/Protobuf, Apache Kafka, BullMQ
Auth & Security: OAuth 2.0 PKCE, JWT, device-bound auth, feature flags
Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, Pulumi, Bazel, Jenkins, GitHub Actions
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, MySQL, SurrealDB, SQLite
AI & GenAI: LangChain, RAG, Qdrant, Pinecone, FAISS, AstraDB, AWS Bedrock

ACHIEVEMENTS

- 500 S&P 500** stock prediction pipeline on **4x NVIDIA H100 GPUs** (DRAC), XGBoost across 26 horizons with warm-start retraining every 15 min. Top **2%** freelancer on Freelancer.com (5-star, 25+ production projects). Won inter-university hackathon.